

THE PROBLEM

- VLMs recognize objects and understand language well
- But fail at physical reasoning: predicting trajectories, inferring collisions, answering counterfactuals
- Root cause: not model capacity. A single frame has no access to depth, motion, or geometry
- The model simply guesses

MY THESIS

A VLM told "blue sphere moving rightward at 1.8 px/frame, depth 0.38, collision at frame 84" should dramatically outperform one given only a raw image.

OUR APPROACH

S1

Object tracking

YOLO + SAM 2

S2

Depth estimation

Depth Anything V2

S3

Optical flow

RAFT-small

S4

Context injection

LLaVA 1.5 7B

No fine-tuning | No new architecture | Pure context injection

Vision-Language Models (VLMs)

Model selection

WHAT ARE VLMs

Multimodal models that take image and text as input and generate text output. Trained on large image-text pairs to understand visual scenes and answer questions.

MY CHOICE

LLaVA-1.5 7B

Large Language and Vision Assistant

[llava-hf/llava-1.5-7b-hf](https://huggingface.co/llava-hf/llava-1.5-7b-hf)



WHY LLaVA

- Single-frame input: isolates our pipeline contribution from the VLM
- Image-native VLMs reduce the measurable impact of context injection
- 4-bit NF4 quantization: fits within A100 compute budget
- Open weights: reproducible, no API costs

OVERVIEW

- CoLLision Events for Video REpresentation and Reasoning
- MIT CSAIL, published at ICLR 2020
- 20,000 synthetic videos of 3D objects colliding on a flat tabletop
- Generated using Bullet physics engine and Blender rendering
- We use 70 validation clips for our experiments

TECHNICAL SPECS

128 frames

per clip

3 shapes

sphere, cube, cylinder

25 fps

5 seconds

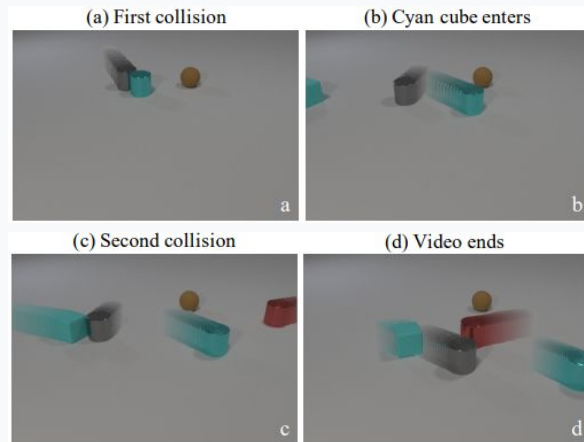
8 colors

fully labeled

480 x 320

resolution

Yi et al. "CLEVRER: Collision Events for Video Representation and Reasoning." ICLR 2020. <http://clevrer.csail.mit.edu>



Experiment: Question Types and Scoring

Descriptive

Open-ended, single word answer

Sample: What color is the first object to collide with the gray sphere?

Explanatory

Multi-select multiple choice

Sample: Which of the following is responsible for the gray cylinder's collision?

Predictive

2-option MC, single correct

Sample: Which event will happen next?

Counterfactual

Multi-select MC, hypothetical

Sample: Without the gray object, which event will not happen?

Experiment: Question Types and Scoring

I. Descriptive

Q: *What shape is the object that collides with the cyan cylinder?*

A: *cylinder*

Q: *How many metal objects are moving when the video ends?*

A: *3*

II. Explanatory

Q: *Which of the following is responsible for the gray cylinder's colliding with the cube?*

a) The presence of the sphere

b) The collision between the gray cylinder and the cyan cylinder

A: *b)*

III. Predictive

Q: *Which event will happen next*

a) The cube collides with the red object

b) The cyan cylinder collides with the red object

A: *a)*

IV. Counterfactual

Q: *Without the gray object, which event will not happen?*

a) The cyan cylinder collides with the sphere

b) The red object and the sphere collide

A: *a), b)*

Per-question accuracy

What it measures: Strictest metric. All choices must match exactly.

How it is calculated: 1 if predicted set equals ground truth set, else 0

Primary metric. Matches official CLEVRER benchmark.

Per-option accuracy

What it measures: Each choice independently scored as binary correct/wrong.

How it is calculated: Fraction of options where prediction matches ground truth label

Primary metric. Matches official CLEVRER benchmark.

F1 score

What it measures: Partial credit metric for multi-select questions.

How it is calculated: $2 \times P \times R / (P + R)$ where P = precision, R = recall over choice sets

Supplementary metric. Edge case: 0 correct answers + model predicts none = F1 of 1.0 (correct behavior).

WHAT THE MODEL RECEIVES

- Last frame of clip (frame 127, JPEG)
- Object identity list: color, material, shape from annotations
- The question and answer format instructions
- ANSWER: priming at end of prompt
- No trajectory, depth, or velocity information

KEY DESIGN DECISIONS

- Last frame used for all question types (full causal chain visible)
- Approach A: full question with all choices in one call
- Tiered parser for answer extraction (3 tiers)
- Vocab normalization: "round" to "sphere", "metallic" to "metal"

PROMPT STRUCTURE

USER: <image>

You are analyzing a single frame from a short video clip...

Objects in this scene:

- Blue rubber sphere
- Red metal cylinder
- Gray rubber cube

Question: Which event will happen next?

- a) The cube collides with the sphere
- b) The cylinder collides with the cube

ANSWER: [single letter only, e.g. a]

REASONING: [2 sentences max]

ANSWER:

Baseline Results

30 questions (directional, not final)

Descriptive

0.318

per-Q

N/A

per-opt

N/A

F1

Temporal counting impossible from one frame

Explanatory

0.000

per-Q

0.533

per-opt

0.292

F1

Near-random per-option; model guessing

Predictive

0.000

per-Q

N/A

per-opt

N/A

F1

Too few samples (1 Q); not yet meaningful

Counterfactual

0.000

per-Q

0.364

per-opt

0.389

F1

Below random; systematically wrong

Key finding: model relies on language priors, not image. Temporal questions are impossible from a single static frame. Poor baseline validates the thesis.

Enhancement Approach: Overview

TO PREDICT COLLISIONS, A MODEL NEEDS TO KNOW:

What objects are in the scene?

Answered by: YOLO object detection

S1

Where are they and how are they moving?

Answered by: SAM 2 tracks objects and detects overlaps provides centroids, entry/exit, visibility

S1

Are objects on the same collision plane?

Answered by: Depth Anything V2 estimates relative depth

S2

How fast and in what direction?

Answered by: RAFT-small estimates optical flow velocity

S3

All measurements injected as natural language context into LLaVA before each question. No fine-tuning, no architectural changes.

Pipeline Architecture: Baseline to Full Enhancement



Stage 1: What Is in the Scene?

Object detection and point prompt generation

WHAT IT ANSWERS

Identifies every object in the scene by color, material, and shape. Provides pixel-accurate centroids to seed SAM 2 tracking.

APPROACHES TRIED

REJECTED

Annotation world coords

3D Bullet physics coords. No projection formula available.

REJECTED

YOLO (v8m, v8l)

0 cube detections. Cylinders misidentified as cups. Spheres partially as sports balls. COCO domain gap.

REJECTED

HSV segmentation

Shadows alter HSV values. Occlusion corrupts blobs. Similar hues cause centroid drift.

SELECTED

CLEVRER proposal file

Mask R-CNN, 0.999+ confidence. Pixel-accurate. CLEVRER-native labels. No domain gap.

MODEL AND IO

Model:

SAM 2 base+ (prompted with proposal centroids)

Input:

128 RGB frames + one centroid point prompt per object

Output:

Per-object centroids (128x2), areas (128), entry/exit frames, overlap pairs

Prompt frame:

first_visible_frame + 15 (capped at 127)

Propagation:

Forward and backward from prompt frame

Storage:

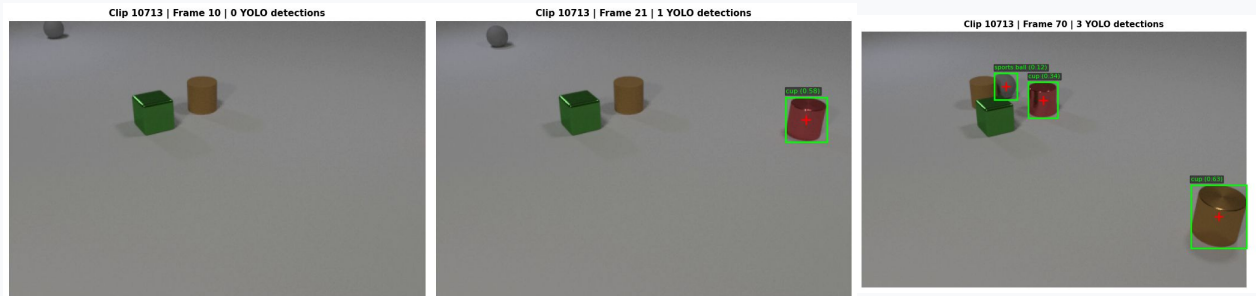
Derived quantities only. Raw masks not saved (60MB/clip).

CONTEXT STRING CONTRIBUTION

Object overlaps detected: 3

Red sphere <> green cylinder: overlap at frame 55 (5px)

Red sphere <> gray sphere: overlap at frame 77 (2px)



```
Clip 10713 -- running YOLO on frames: [10, 21, 70]
  Frame 10: 0 detections -- []
  Frame 21: 1 detections -- ['cup']
  Frame 70: 3 detections -- ['cup', 'cup', 'sports ball']
Hence, NO YOLO!
```

Using author provided 'proposal' files

```
Clip 10205 -- detected 4 objects:
  red rubber sphere -> centroid (224.7, 91.7) at frame 10
  green metal cylinder -> centroid (427.9, 158.6) at frame 24
  gray rubber sphere -> centroid (36.5, 177.1) at frame 45
  brown rubber cube -> centroid (22.2, 144.1) at frame 81
```

Stage 2: Where Are the Objects and How Do They Move?

SAM 2 video object segmentation

WHAT IT ANSWERS

Tracks every object across all 128 frames. Provides pixel-level masks, centroid trajectories, entry/exit events, and mask overlap detection.

DESIGN DECISIONS

- SAM 2 over YOLO: pixel masks enable far more precise measurements than bounding boxes
- Prompt frame: `first_visible_frame + 15`, not frame 0 (allows object to fully enter scene)
- Overlap detection: pixel-level mask intersection across all 128 frames, first instance only
- Language: "masks overlap at frame X" not "collision at frame X". Let LLaVA decide.
- Overlap pixel counts small (1-7px): confirms precision of detection

LIMITATION

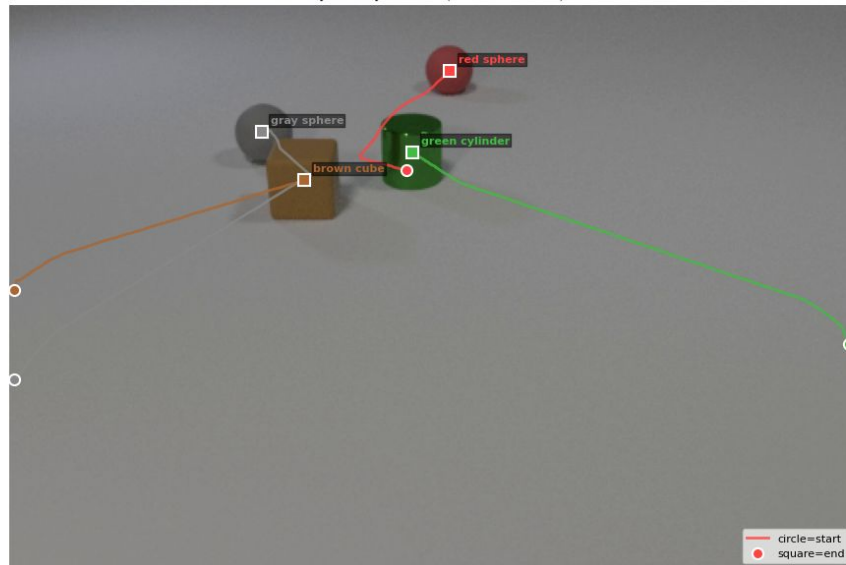
Raw masks not saved (60MB/clip exceeds Drive budget). Downstream stages use centroid window approximation instead of full mask averaging.

INPUT / OUTPUT

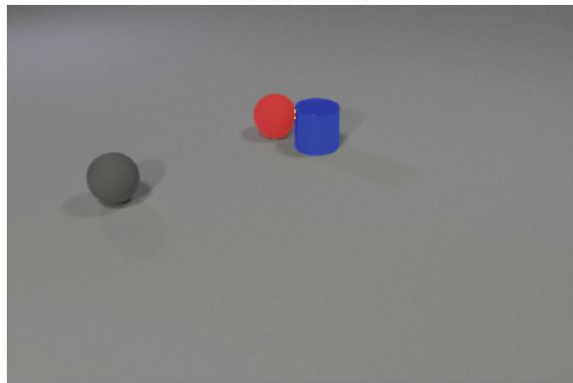
Input:	128 RGB frames + centroid point prompts from proposal file
Output:	Centroids (128x2 float), areas (128), entry frame, exit frame, overlap pairs with frame and pixel count
Model:	SAM 2 base+ (0.65GB VRAM on A100)
Saved:	stage1_sam2_{vid_id:05d}.npz, compressed

Clip 10205 -- Trajectory + Overlap Sanity Check

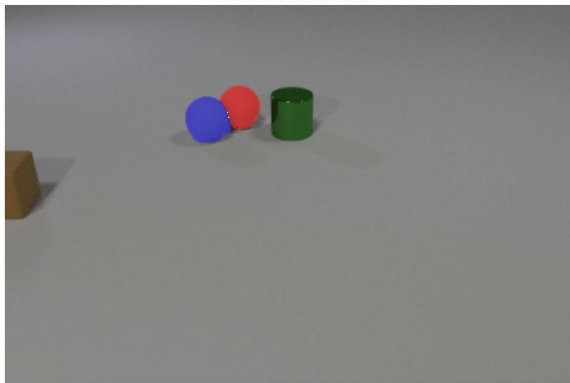
Object Trajectories (all 128 frames)



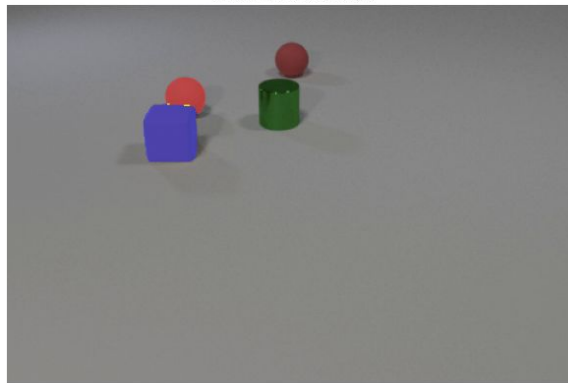
red sphere (red) <-> green cylinder (blue)
First overlap frame 55 (5 px)



red sphere (red) <-> gray sphere (blue)
First overlap frame 77 (2 px)



gray sphere (red) <-> brown cube (blue)
First overlap frame 115 (7 px)



Stage 3: Are Objects on the Same Collision Plane?

Depth Anything V2 Small

WHAT IT ANSWERS

Establishes whether two overlapping objects are physically co-planar, or whether the overlap is a projection artifact with no physical contact.

DESIGN DECISIONS

- DAV2-Small (not Large): sufficient for coarse depth on synthetic flat tabletop
- Frame sampling: overlap frames + last frame only (not 8 uniform frames)
- Depth is relative within each frame only. Cross-frame comparison is invalid.
- Collision-plane threshold 0.05: empirically derived from ground truth annotation collisions at t-5 frames
- Final scene section: depth snapshot at frame 127 directly supports predictive/counterfactual reasoning

INPUT / OUTPUT

Model:

Depth Anything V2 Small (HuggingFace pipeline)

Input:

Single RGB frame per inference call

Output:

Depth scalar per object at overlap frames and frame 127

Threshold:

0.05 depth difference = same collision plane

Saved:

stage2_depth_{vid_id:05d}.npz

Depth at overlap frame 55: red sphere 0.39, green cylinder 0.41, diff 0.02 (same collision plane).
Final scene depth: red sphere 0.16, green cylinder 0.35, gray sphere 0.33, brown cube 0.40

CONTEXT STRING CONTRIBUTION

Stage 3: Demo

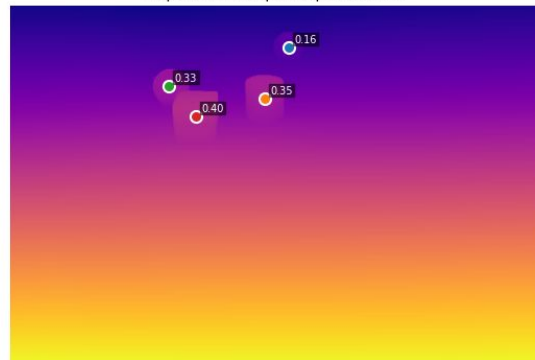
Depth maps and per-object depth values

DAv2 Depth Sanity Check -- Last Frame (Frame 127)

Clip 10205 -- RGB Frame 127



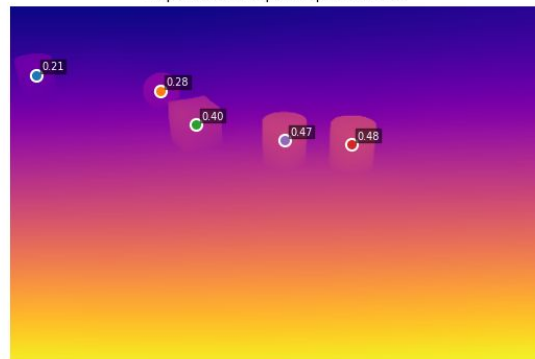
Clip 10205 -- Depth Map Frame 127



Clip 10713 -- RGB Frame 127



Clip 10713 -- Depth Map Frame 127



Stage 4: How Fast and In What Direction?

RAFT-small optical flow

WHAT IT ANSWERS

Provides the only temporal signal in the pipeline. Velocity vectors tell LLaVA which objects are moving, how fast, and in what direction before and after collisions.

DESIGN DECISIONS

- RAFT-small (not Large): sufficient for synthetic clean-edge solid-color objects
- 3-frame gap: (overlap_frame - 3, overlap_frame) and (124, 127). 1-frame gap too small.
- 15x15 centroid window: raw masks not saved, window approximates mask averaging
- Two velocity sections: pre-overlap pairs and final velocities
- Direction as natural language: "moving rightward and upward (+1.81, -0.24)"

CONTEXT STRING CONTRIBUTION

Velocity before overlap at frame 55:
green cylinder moving leftward and upward (-18.61, -3.41)
Final: red sphere moving rightward (+3.10, -5.42)

INPUT / OUTPUT AND LIMITATIONS

Model:

RAFT-small (torchvision, GPU required)

Input:

Two consecutive RGB frames, float32 in [0, 255]

Output:

Velocity (dx, dy) per object at overlap frames and frame 124-127

Saved:

stage3_raft_{vid_id:05d}.npz

Limitations:

- 15x15 window samples background pixels for small objects
- Centroid drift at occlusion frames: unreliable vectors
- Collision frame instability: e.g. dx = -49.76 outlier observed

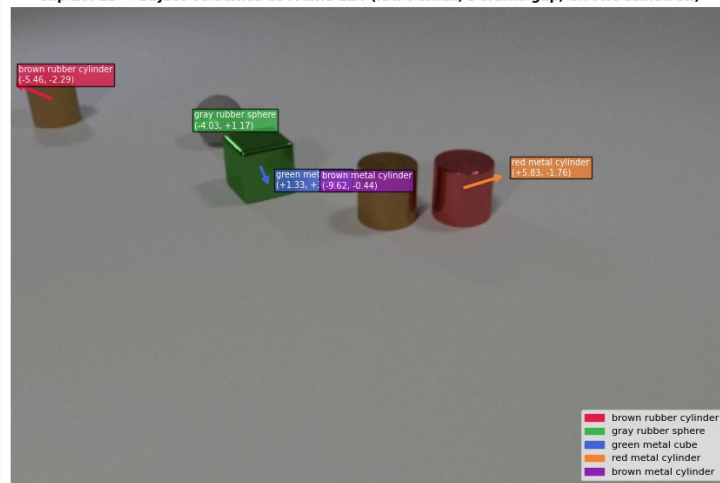
Stage 4: Demo

Optical flow arrows and velocity vectors

Clip 10205 -- Object Velocities at Frame 124 (RAFT-small, 3-frame gap, arrows scaled 5x)



Clip 10713 -- Object Velocities at Frame 124 (RAFT-small, 3-frame gap, arrows scaled 5x)



— SAM2 context preview (clip 10205) —————

Objects detected in this video clip:

- Red rubber sphere: enters scene at frame 15, visible for 113/128 frames.
- Green metal cylinder: enters scene at frame 15, visible for 113/128 frames.
- Gray rubber sphere: enters scene at frame 36, visible for 92/128 frames.
- Brown rubber cube: enters scene at frame 72, visible for 56/128 frames.

Object overlaps detected: 3

red sphere <-> green cylinder first overlap at frame 55 (5 pixels)

red sphere <-> gray sphere first overlap at frame 77 (2 pixels)

gray sphere <-> brown cube first overlap at frame 115 (7 pixels)

— Depth context preview (clip 10205) —————

Depth information (relative estimates within each frame,

0 = closest to camera, 1 = farthest from camera).

Objects with depth difference below 0.05 are likely on the same collision plane.

Depth at spatial overlap events:

- Red rubber sphere and green metal cylinder at frame 55: depths 0.39 and 0.42, difference 0.04 (same collision plane).
- Red rubber sphere and gray rubber sphere at frame 77: depths 0.34 and 0.38, difference 0.04 (same collision plane).
- Gray rubber sphere and brown rubber cube at frame 115: depths 0.31 and 0.42, difference 0.11 (different planes).

Final scene depth (frame 127):

- Brown rubber cube: depth 0.40
- Gray rubber sphere: depth 0.33
- Green metal cylinder: depth 0.35
- Red rubber sphere: depth 0.16

— RAFT context preview (clip 10205) —————

Velocity information (pixels per frame, 3-frame gap):

Positive x = rightward, negative x = leftward.

Positive y = downward, negative y = upward.

Note: velocity estimates are approximate. Readings at some overlap frames may be less reliable due to occlusion.

Velocity just before each spatial overlap event:

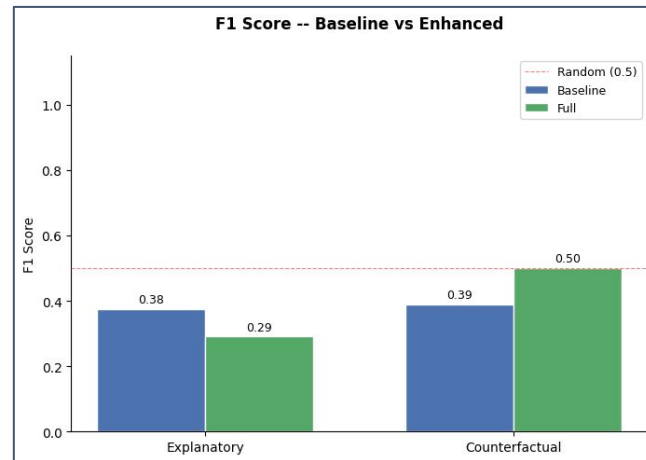
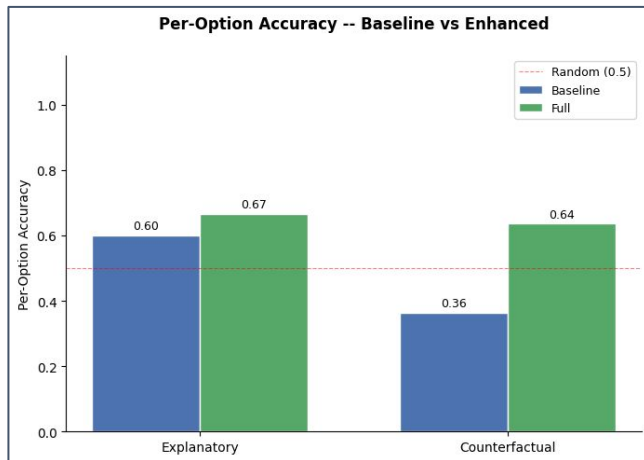
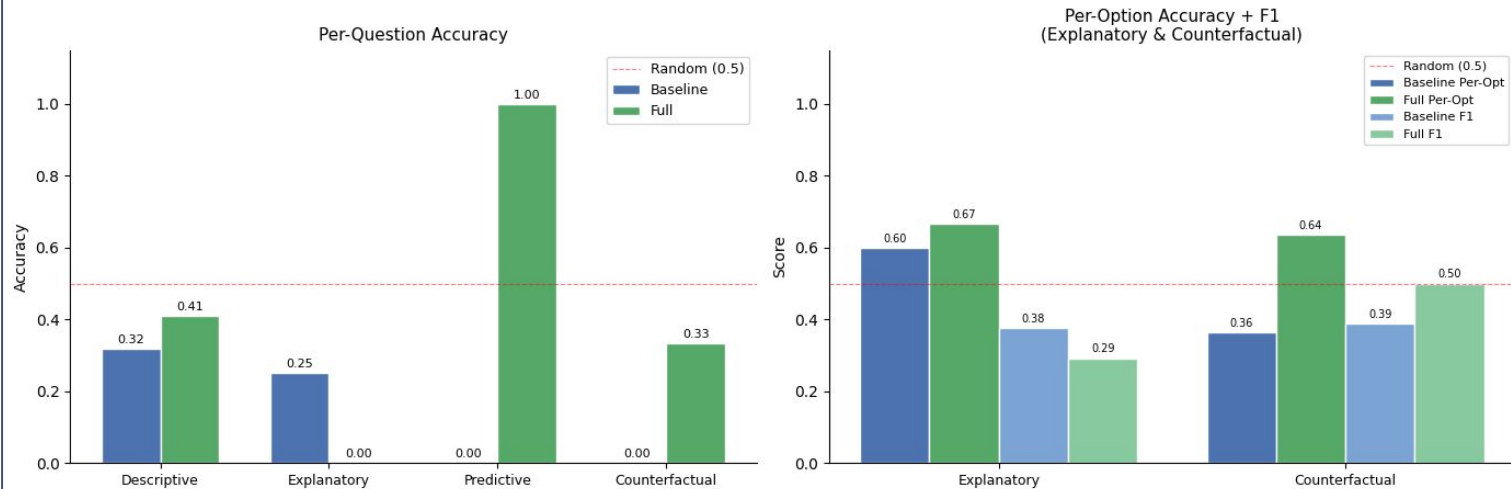
- Overlap at frame 55: Red rubber sphere moving nearly stationary (+0.28, -0.24), green metal cylinder moving leftward and upward (-18.61, -3.41).
- Overlap at frame 77: Red rubber sphere moving leftward and upward (-5.94, -1.86), gray rubber sphere moving rightward and upward (+8.13, -5.88).
- Overlap at frame 115: Gray rubber sphere moving rightward and upward (+2.26, -4.83), brown rubber cube moving rightward and upward (+8.44, -3.51).

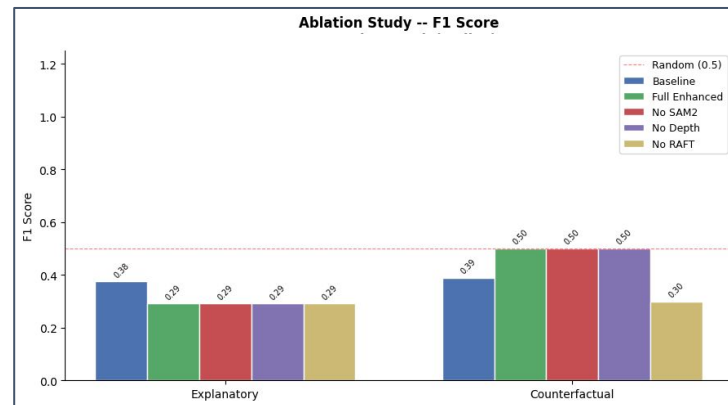
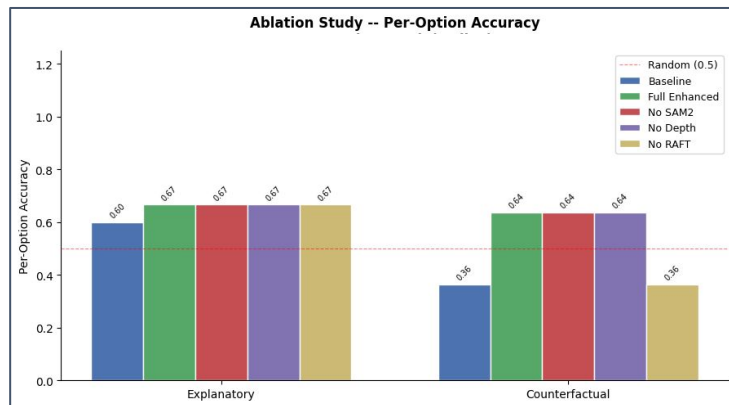
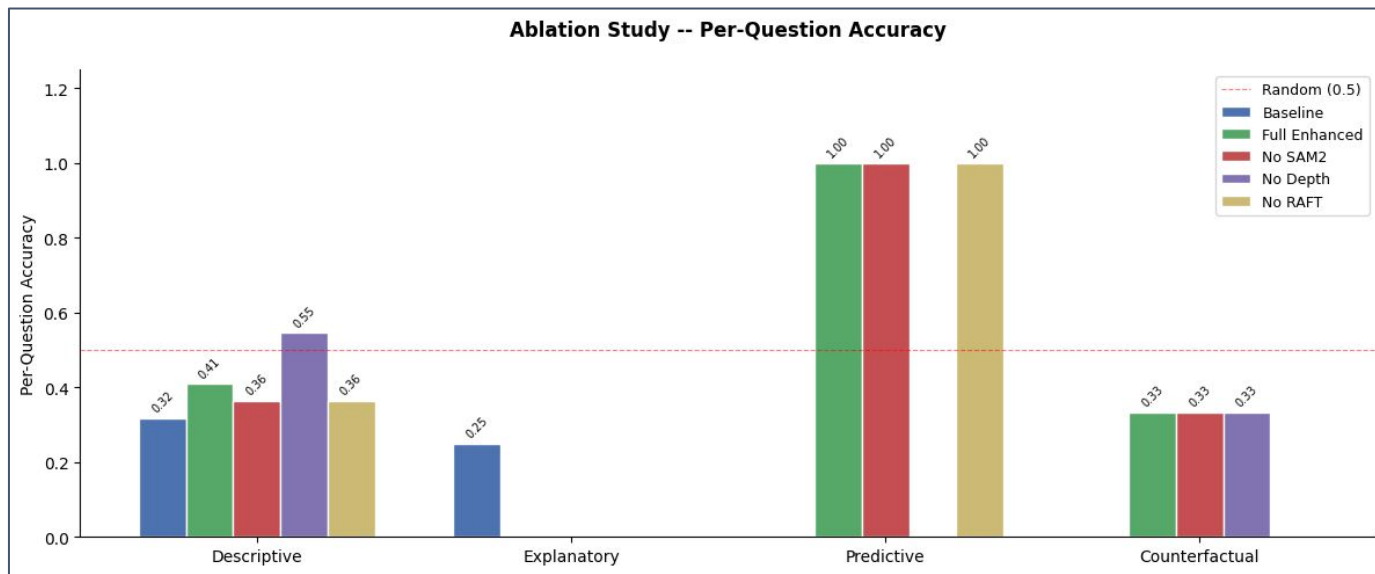
Final object velocities (frames 124 to 127):

- Red rubber sphere: moving rightward and upward (+3.10, -5.42).
- Green metal cylinder: moving leftward and downward (-3.34, +0.35).
- Gray rubber sphere: moving rightward and upward (+1.89, -0.99).
- Brown rubber cube: moving rightward (+6.12, -0.20).

Type	Condition	Total Q	Per-Q Acc	Per-Opt	F1
descriptive	baseline	22	0.3182	--	--
descriptive	full	22	0.4091	--	--
descriptive	no_sam2	22	0.3636	--	--
descriptive	no_depth	22	0.5455	--	--
descriptive	no_raft	22	0.3636	--	--
explanatory	baseline	4	0.2500	0.6000	0.3750
explanatory	full	4	0.0000	0.6667	0.2917
explanatory	no_sam2	4	0.0000	0.6667	0.2917
explanatory	no_depth	4	0.0000	0.6667	0.2917
explanatory	no_raft	4	0.0000	0.6667	0.2917
predictive	baseline	1	0.0000	--	--
predictive	full	1	1.0000	--	--
predictive	no_sam2	1	1.0000	--	--
predictive	no_depth	1	0.0000	--	--
predictive	no_raft	1	1.0000	--	--
counterfactual	baseline	3	0.0000	0.3636	0.3889
counterfactual	full	3	0.3333	0.6364	0.5000
counterfactual	no_sam2	3	0.3333	0.6364	0.5000
counterfactual	no_depth	3	0.3333	0.6364	0.5000
counterfactual	no_raft	3	0.0000	0.3636	0.3000

PhysicsStack Results





Final Results: Scores vs Baseline

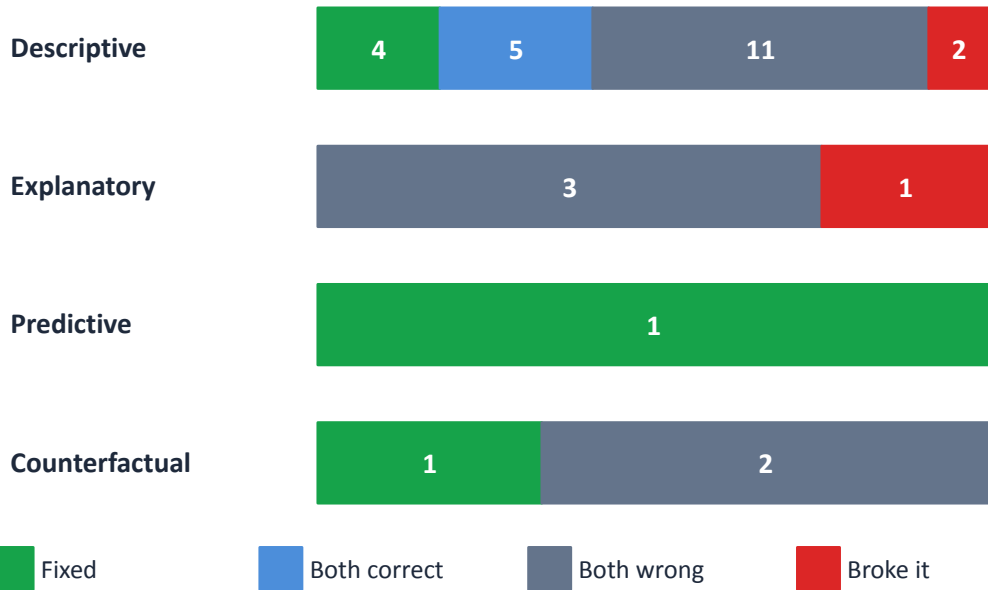
2 clips, 30 questions (scaling to 70 in progress)

Question Type	Per-Q Baseline	Per-Q Full	Per-Opt Baseline	Per-Opt Full	F1 Baseline	F1 Full
Descriptive	0.32	0.41	N/A	N/A	N/A	N/A
Explanatory	0.25	0.22	0.53	0.35	0.29	0.22
Predictive	0.00	1.00	N/A	N/A	N/A	N/A
Counterfactual	0.00	0.33	0.36	0.64	0.39	0.50

Predictive: 0.00 to 1.00. Depth is the critical signal (no_depth condition drops back to 0.00). Explanatory regression: context confuses causal chain.

Final Results: Analysis and Key Findings

CONFUSION BREAKDOWN (BASELINE vs FULL)



Pipeline fixed 6 questions, broke 3. Net positive.

Context IS changing LLaVA behavior. But LLaVA does keyword extraction, not physics reasoning. "Same collision plane" gets latched onto and misapplied.

CRITICAL FAILURE CASE

The 1.89 velocity leak

Shape question answered with "1.89" (a RAFT velocity value). RAFT numerical measurements leaked into the shape answer field.

Shows risk of injecting raw numerical context into VLMs not trained to handle structured measurements.

ABLATION FINDINGS

- Depth most critical: removing it drops predictive from 1.0 to 0.0
- SAM 2 alone: marginal improvement on descriptive
- RAFT: improves directional reasoning but introduces outlier risk
- Full context: net positive across all types except explanatory

Future Work

Scale evaluation

Run full pipeline on all 500 CLEVRER validation clips for statistically meaningful results. Current 2-clip results are directional signals only.

Full mask flow sampling

Save raw SAM 2 masks to enable full mask averaging in RAFT. Replace 15x15 centroid window approximation with precise object-region flow.

Video-native VLM comparison

Add Video-LLaVA baseline to compare implicit temporal understanding vs explicit structured context injection.

Real-time demo

Replace pipeline components with speed-optimized variants: YOLOv8n, DAv2-Small, RAFT-base, Moondream2. Target 15fps on A100.

LLaVA fine-tuning

Fine-tune LLaVA on structured physical context format to reduce keyword extraction failures and numerical value leakage.

Learnings

+

Structured symbolic context can meaningfully improve VLM physical reasoning without any fine-tuning

+

Depth is the single most critical signal for collision prediction. Without it, predictive accuracy collapses.

!

Domain gap is a real and severe problem. COCO-trained YOLO completely fails on synthetic geometric objects.

!

Context injection is not universally beneficial. Over-injecting numerical measurements causes format confusion and value leakage.

x

VLMs do keyword extraction, not physics reasoning. "Same collision plane" gets latched onto and misapplied to unrelated answers.

+

Clean controlled experiments matter. Object identity must be in both conditions for fair comparison.